TOSAM-JESSE WINTER

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PROFILE

MSc Robotics student at the University of Manchester and First-Class Electrical and Electronic Engineering graduate (CGPA: 5.00/5.00; Class of 2025 Valedictorian). Experience spanning robotics, computer vision, Python-based machine learning, control systems and electrical power engineering. Published researcher with hands-on engineering project and substation experience.

EDUCATION

The University of Manchester | September 2026 – Present

- MSc Robotics
- Relevant Coursework: Software for Robotics, Cognitive Robotics and Computer Vision, Robotic Systems, Robotic System Design Project

Nile University of Nigeria | October 2020 – November 2025

- Bachelor of Electrical and Electronic Engineering, First-Class Honours (**CGPA**: 5.00/5.00)
- Overall Best Graduating Student (Valedictorian), Class of 2025; Best Graduating Student in the Department and Faculty of Engineering.
- Relevant Coursework: Electronics, Computational Intelligence, Digital Signal Processing, Control Engineering, Digital Control Systems, Digital Communications, Data Communication and Networks

WORK EXPERIENCE

Nile University of Nigeria | Teaching and Administrative Assistant | June 2026 – September 2026

- Supported undergraduate teaching in the Department of Electrical and Electronic Engineering, assisting academic staff with lecture delivery.
- Undertook departmental administrative activities and contributed to teaching-related coordination and communication.

Transmission Company of Nigeria (TCN) | March 2024 – September 2025

- Worked as a student intern at the Apo 132kV/33kV substation as part of the Students Industrial Work Experience Program (SIWES)
- Participated in transformer commissioning/decommissioning, instrument transformer testing, and relay protection system upgrades.
- Assisted in substation preventive and corrective maintenance, fault detection, and transmission line maintenance.

TECHNICAL SKILLS

- **Programming and analysis:** Python (NumPy, Pandas, Matplotlib, scikit-learn), MATLAB and Simulink.
- **Robotics and software:** ROS 2, RViz, Linux, Git and VS Code.
- **AI and computer vision:** TensorFlow, Keras, OpenCV, image classification and transfer learning.
- **Engineering and productivity:** AutoCAD (Electrical), protection systems, technical documentation and Microsoft Office.

PROJECTS

LeoRover – Supermarket Item Retrieval (Group Project) | September 2026 – Present

- Contributing to a ROS 2-based mobile rover project for identifying and retrieving colour-coded objects in a simulated supermarket setting.
- Worked with robot network configuration, 2D LiDAR scan data, RViz visualisation and ROS 2 software environments.

Autonomous Herbicide Spraying Drone | November 2024 – June 2025

- Designed and constructed an agricultural drone prototype combining camera-based crop inspection with an electronically controlled spraying mechanism.
- Developed a Raspberry Pi computer-vision workflow using Python, OpenCV and machine learning; integrated image-based detection with pump actuation.

Plant Disease Detection System | February 2025 – April 2025

- Developed AI-based image recognition of various plant diseases from different crops for real-time crop health monitoring.

Breast Cancer Detection System | April 2025 – May 2025

- Implemented a machine learning model for early detection and classification of breast tumours from medical imaging datasets.

PUBLICATIONS

Breast Cancer Classification Using Machine Learning Techniques and Convolutional Neural Networks | October 2025

- Published in the Nile Journal of Engineering and Applied Sciences, 3(1), 521–534 (2025). DOI: 10.5455/NJEAS.256827.
- Investigated machine learning methods and convolutional neural networks for breast cancer image classification.

LEADERSHIP & ACTIVITIES

- Secretary General, Nigerian Universities Engineering Students Association (NUESA) Secretary General | January 2022 – February 2023
- Student Member, Institute of Electrical and Electronic Engineers (IEEE) | February 2025 – December 2025
- Student Member, Nigerian Institute of Electrical and Electronic Engineers (NIEEE), Nile Student Chapter | November 2021 – November 2025
- Toastmasters Leadership and Public Speaking Training | July 2019 – August 2019